



END SEMESTER ASSESSMENT (ESA)
B.TECH. (CSE)
III SEMESTER
UE23CS251A – DIGITAL DESIGN & COMPUTER ORGANIZATION
LABORATORY

DDCO Mini-Project

Design and implement a 4 bit register using 4 d flip flops and 4 multiplexers and 2 mode selection lines s1 and s0.

Section: A

Team Number: 17

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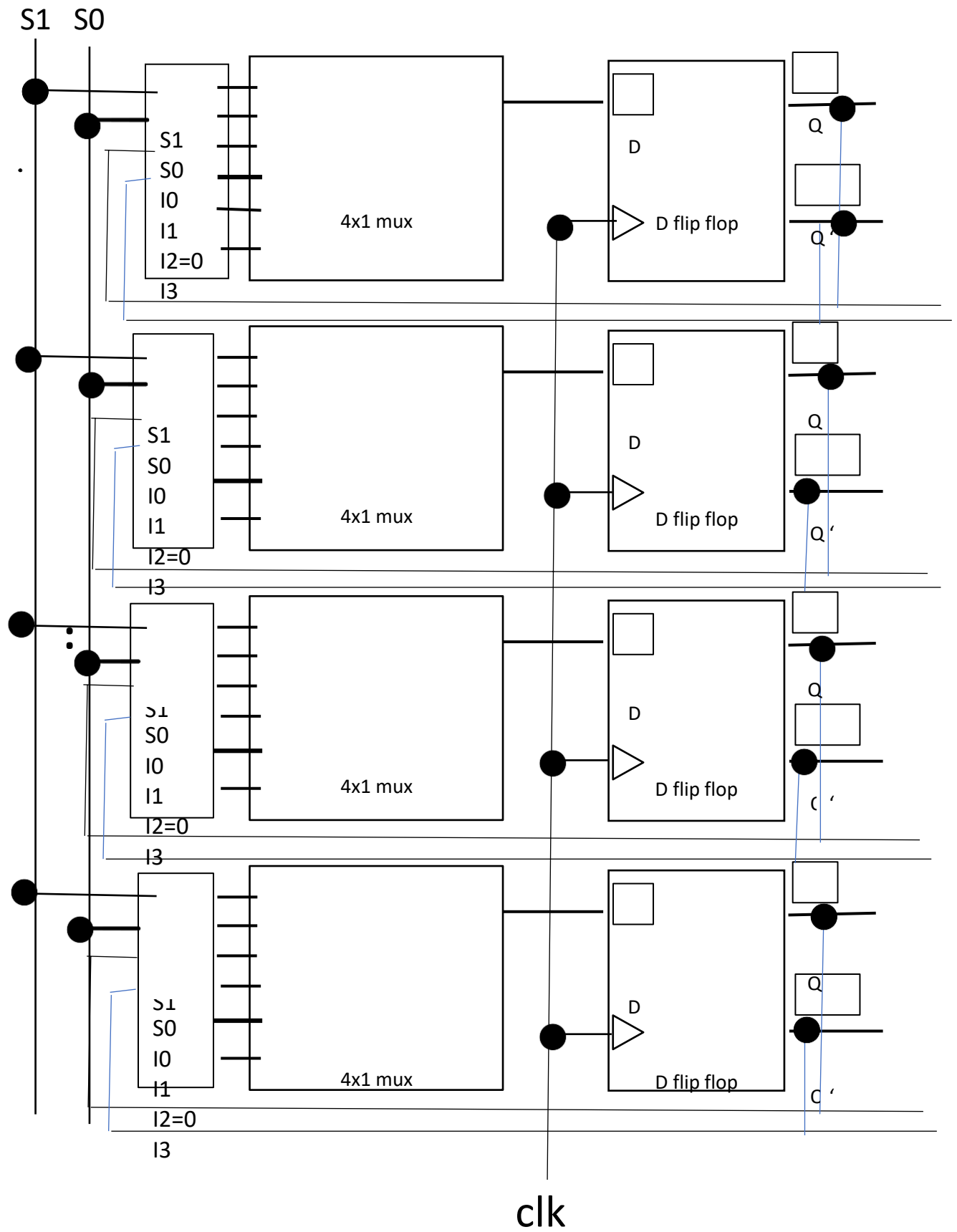
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TABLE OF CONTENTS		
Sl.No	TOPIC	PAGE No
1.	ABSTRACT OF THE PROJECT	<u>3</u>
2.	CIRCUIT DIAGRAM	<u>4</u>
3.	MAIN VERILOG CODE	<u>5</u>
4.	TEST BENCH FILE	<u>7</u>
5.	SCREEN SHOTS OF THE OUTPUT	<u>10</u>

ABSTRACT OF THE PROJECT:

This project focuses on the design and implementation of a 4-bit register using four D flip-flops and four 4×1 multiplexers, controlled by mode selection inputs s_1 and s_0 . The register performs 4 distinct operations based on these inputs: it holds the current value if $s_1s_0 = 00$, complements (inverts) the output when $s_1s_0 = 01$, shifts all bits to the right if $s_1s_0 = 10$, and shifts all bits to the left when $s_1s_0 = 11$. This design highlights how digital circuits utilize sequential and combinational logic to achieve versatile data storage and manipulation, offering insights into core concepts of digital system design and computer architecture.

CIRCUIT DIAGRAM:



MAIN VERILOG CODE:

```
module FourBitRegister (  
  
    input wire [3:0] D,  
  
    input wire [1:0] S,  
  
    input wire clk,  
  
    input wire reset,  
  
    output reg [3:0] Q  
  
);  
  
    reg [3:0] next_state;  
  
    always @(*) begin  
  
        case (S)  
  
            2'b00: next_state = D;  
  
            2'b01: next_state = ~Q;
```

```
2'b10: next_state = {1'b0, Q[3:1]};
```

```
2'b11: next_state = {Q[2:0], 1'b0};
```

```
default: next_state = Q;
```

```
endcase
```

```
end
```

```
always @(posedge clk or posedge reset) begin
```

```
    if (reset)
```

```
        Q <= 4'b0000;
```

```
    else
```

```
        Q <= next_state;
```

```
end
```

```
endmodule
```

TEST BENCH FILE:

```
`timescale 1ns / 1ps

module FourBitRegister_tb;

    reg [3:0] D;
    reg [1:0] S;
    reg clk;
    reg reset;

    wire [3:0] Q;

    FourBitRegister uut (
        .D(D),
        .S(S),
        .clk(clk),
        .reset(reset),
        .Q(Q)
    );

    initial begin
        clk = 0;
        forever #5 clk = ~clk;
    end

    initial begin
        D = 4'b0000;
        S = 2'b00;
```

```
reset = 1;
```

```
@(posedge clk) reset = 0;
```

```
@(posedge clk) begin  
    S = 2'b00;  
    D = 4'b1010;  
end
```

```
@(posedge clk) begin  
    S = 2'b01;  
end
```

```
@(posedge clk) begin  
    S = 2'b10;  
end
```

```
@(posedge clk) begin  
    S = 2'b11;  
end
```

```
@(posedge clk) begin  
    reset = 1;  
end
```

```
@(posedge clk) reset = 0;
```

```
@(posedge clk) begin  
    S = 2'b00;  
    D = 4'b1111;  
end
```


end

```
@(posedge clk) begin
    S = 2'b10;
    D = 4'b1100;
end
```

```
@(posedge clk) begin
    S = 2'b11;
    D = 4'b0011;
end
```

```
#20 $finish;
end
```

```
initial begin
    $monitor("Time = %0t | S = %b | D = %b | Q
= %b | reset = %b",
            $time, S, D, Q, reset);
end
```

```
initial begin
    $dumpfile("FourBit.vcd");
    $dumpvars(0, FourBitRegister_tb);
end
```

endmodule

SCREEN SHOTS OF THE OUTPUT:

```
C:\iverilog\bin>iverilog -o test miniproject[1].v miniproject_tb[1].v

C:\iverilog\bin>vvp test
VCD info: dumpfile FourBit.vcd opened for output.
Time = 0 | S = 00 | D = 0000 | Q = 0000 | reset = 1
Time = 5000 | S = 00 | D = 0000 | Q = 0000 | reset = 0
Time = 15000 | S = 00 | D = 1010 | Q = 0000 | reset = 0
Time = 25000 | S = 01 | D = 1010 | Q = 1010 | reset = 0
Time = 35000 | S = 10 | D = 1010 | Q = 0101 | reset = 0
Time = 45000 | S = 11 | D = 1010 | Q = 0010 | reset = 0
Time = 55000 | S = 11 | D = 1010 | Q = 0000 | reset = 1
Time = 65000 | S = 11 | D = 1010 | Q = 0000 | reset = 0
Time = 75000 | S = 00 | D = 1111 | Q = 0000 | reset = 0
Time = 85000 | S = 10 | D = 1100 | Q = 1111 | reset = 0
Time = 95000 | S = 11 | D = 0011 | Q = 0111 | reset = 0
Time = 105000 | S = 11 | D = 0011 | Q = 1110 | reset = 0
miniproject_tb[1].v:69: $finish called at 115000 (1ps)
Time = 115000 | S = 11 | D = 0011 | Q = 1100 | reset = 0

C:\iverilog\bin>gtkwave FourBit.vcd

GTKWave Analyzer v3.3.100 (w)1999-2019 BSI

[0] start time.
[115000] end time.
```

